

General Industries Hub

Enterprise Inventory Management

Standard Operating Procedure - Procurement Chain

Version 3.0 - Generated 2026-06-17

Confidential - for authorized personnel only

GI Hub - Streamlit + SQLite + Twilio + Ollama - Multi-Site ERP

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General Industries Hub - Standard Operating Procedure

Version 1.0 - Procurement Chain (Logistics + Warehouse + Site) Owners: Logistics Manager - Warehouse Lead - Site HOD Council Applies to: All sites + warehouses connected to the GI Hub ERP Companion docs: USER_MANUAL.md (every page/tab/button) - handoff.md (technical architecture) Cadence: Review quarterly; update on any role change, vendor onboarding, or DN-state-machine alteration.

§1. Purpose & Scope

This SOP turns the procurement-chain features of GI Hub v3.0 into a daily / weekly operating rhythm. It answers *"what should I do at 8 AM Tuesday?"* - for that, USER_MANUAL is too granular and handoff.md too technical.

In scope:

- Daily / weekly / monthly cadences for Site HOD, Logistics, Warehouse User, Store Keeper, Admin
- Decision trees for the four most common judgment calls (reject vs reschedule vs force-close, return vs adjustment, override timing, when to close)
- Escalation matrix - who-pings-who-after-how-long, with WhatsApp / email / admin override
- Recovery procedures - DN rejected at HOD, warehouse short-receives, vendor return + resupply, HOD absent
- Quick reference cards per role

Out of scope:

- UI screenshots (use USER_MANUAL.md)
- Code / SQL details (use handoff.md)
- Hosting + backup procedures (use handoff.md §4 and USER_MANUAL §17)

§2. Role Responsibility Matrix (RACI)

R = Responsible (does the work) - A = Accountable (signs off) - C = Consulted (input) - I = Informed (notified)

Activity	Site HOD	Logistics	Warehouse User	Store Keeper	Admin
Create PR	R/A	I	-	-	I
Submit PR to Logistics	R/A	I	-	-	I
Negotiate vendor + price	I	R/A	-	-	C

Activity	Site HOD	Logistics	Warehouse User	Store Keeper	Admin
Issue PO	I	R/A	-	-	I
Upload PO PDF	-	R/A	-	-	-
Assign PO to a warehouse	I	R/A	I	-	I
Acknowledge assignment	-	I	R/A	-	-
Receive goods from vendor	-	I	R/A	-	-
Prepare DN (RL/BL)	-	I	R/A	-	-
Approve DN delivery date	-	R/A	I	-	I
Approve DN content	R/A	I	I	-	I
Mark DN as received (site)	I	I	I	R/A	-
Request reschedule	C/R	A	C/R	-	I
Raise vendor return	C	A	C/R	C	I
Force-close PR / PO / line	I	R/A	-	-	I (critical)
Approve user registrations	-	-	-	-	R/A
Maintain vendor / warehouse master	-	C	C	-	R/A
Review audit logs	-	-	-	-	R/A
Resolve oversight	C	C	C	C	R/A

escalations

§3. Cadence Checklists

§3.1 Site HOD - Daily

Morning (8:00 AM):

- [] Open HOD Portal -> check the bell badge in the sidebar
- [] Open inbox -> mark non-critical items read; act on critical (red border)
- [] EOD Commit tab - confirm yesterday's commits posted cleanly (no pending rows from yesterday)
- [] ? Pending Receipts - review and approve any inbound receipts (legacy path)
- [] ? DN Approvals (NEW) - approve or reject any DN's the Warehouse has prepared

-

☐ ? In-Transit -> "Active in-transit" sub-tab - scan ETAs for the day

Through the day:

- ☐ When PRs are raised - submit to Logistics within 1 hour (? Submit PR(s) to Logistics expander)
- ☐ Reschedules from team - verify before submitting, include real reason
- ☐ EOD Commit by 17:00 - typed COMMIT confirmation

Evening (before sign-off):

- ☐ ? In-Transit -> confirm tomorrow's expected DNs are still on schedule
- ☐ Force-closures sub-tab - anything new since last login?
- ☐ Sign out cleanly (audit log)

§3.2 Site HOD - Weekly (Sunday or first workday)

- ☐ Purchase Requests - clean up any PR in site_draft longer than 7 days (either submit or delete)
- ☐ ? In-Transit -> My reschedule requests - chase any "pending" older than 48 hours
- ☐ Force-closures - review the past 7 days; raise an admin ticket on any without clear reason
- ☐ Cross-Site requests - fulfill outstanding incoming requests

§3.3 Site HOD - Monthly

- ☐ PR backlog review - anything still in submitted/in_po from > 30 days ago needs a call to Logistics
- ☐ Generate ? PO Status report scoped to your site - review on-time delivery rate
- ☐ Audit your own bell inbox for missed criticals (filter to unread)

§3.4 Logistics - Hourly (during work hours)

- ☐ bell badge - clear unread within 30 minutes during business hours
- ☐ Incoming PRs queue - pick up new PRs within 4 hours; acknowledge to Site HOD via WhatsApp if delayed
- ☐ ? Reschedules - decide within 2 hours during business hours; out-of-hours, EOD next business day

§3.5 Logistics - Daily

Morning (8:00 AM):

- ☐ Run the day's PO PDF backlog (any vendor PDFs sitting in your email)
- ☐ Open POs -> filter to "Expected" = today + tomorrow -> confirm each has a warehouse assignment
- ☐ Open POs -> filter to "Expected" = yesterday -> if not delivered, chase warehouse + vendor
- ☐ ?? DN approval queue - approve same-day if delivery date is reasonable
- ☐ <-? Vendor Returns - chase open returns > 7 days old

End of day:

- ☐ Confirm every PO assigned today shows in the receiving Warehouse's notification log

-

[] T-1 dashboard (Admin Portal -> Logistics Oversight -> DNs sub-tab) - pre-empt tomorrow's deliveries

§3.6 Logistics - Weekly

- [] Review the WHATSAPP_TRIGGERS toggles in config.py - adjust if noise complaints came in
- [] Generate ? Warehouse Throughput report - flag underperforming warehouses to Warehouse Lead
- [] Generate ? Force-Closures report - review the week's closures with Admin
- [] Vendor performance review - open returns + delivery delays per vendor

§3.7 Logistics - Monthly

- [] Vendor master cleanup - deactivate dormant vendors (>90 days no PO)
- [] PR-to-PO cycle-time analysis - is the chain getting faster or slower?
- [] Force-closure trend - too many means upstream PR quality is poor

§3.8 Warehouse User - Per Shift

Shift start:

- [] bell badge - clear any unread assignments / DN rejections
- [] Incoming Assignments tab - [OK] Acknowledge new assignments within 1 hour
- [] Review today's Expected_Delivery list - physically prepare receiving area

During shift (when goods arrive):

- [] Receive Goods -> record actual qty per line as physical inspection completes
- [] If short-receive vs PO qty - note the discrepancy in remarks; Logistics will be notified
- [] If quality issue at receive - DON'T add to stock; raise via <-? Returns from Site or a vendor-return note to Logistics

Before DN preparation:

- [] Confirm RL and BL items go on SEPARATE DNs (system enforces, but plan physically too)
- [] Verify destination site for each DN
- [] Driver + vehicle details ready

Shift end:

- [] ?? Outbound DNs -> confirm every DN drafted today is submitted to Logistics (no orphan drafts)
- [] Sign off - audit-logged

§3.9 Warehouse User - Weekly

- [] History tab - sanity-check completed DNs match physical movement records
- [] Open assignments older than 7 days - chase Logistics for closure or vendor follow-up
- [] RL/BL inventory walk - physically verify families are separated in the warehouse

§3.10 Warehouse User - Monthly

-

- [] Throughput review with Warehouse Lead - DNs prepared, on-time rate, rejection rate
 - [] Vehicle / driver master refresh - update Vehicle_No / Driver_Name defaults
-

§3.11 Store Keeper - Each Delivery Day

- [] Entry Log -> Receipt Staging - the ? Incoming Delivery Notes expander shows pending DNs
- [] Physical truck arrives -> inspect, count, match against DN
- [] If everything matches -> [OK] Mark as Received on the DN card
- [] If discrepancy -> tell HOD before marking; HOD escalates to Logistics

§3.12 Store Keeper - Standard Daily (unchanged from v2.0)

See USER_MANUAL.md §4 for the Consumption Log / Receipt Staging / Return Items / Returnable Items / Stock Count / QR Label Request daily routines.

§3.13 Admin - Weekly

- [] Admin Portal -> ? Logistics Oversight -> review every sub-tab
- [] Force-closures sub-tab - any not flagged "agreed-by-stakeholders" -> ask Logistics
- [] ? Users -> review pending registrations (Logistics + Warehouse users always go through admin approval)
- [] ? Audit Logs -> filter by FORCE_CLOSE_* actions

§3.14 Admin - Monthly

- [] Generate all 3 procurement reports (PO Status, Warehouse Throughput, Force-Closures)
- [] Bell-noise audit - check app_notifications row count vs prior month; sudden spike means a process change is needed
- [] Vendor + Warehouse master review - deactivate stale entries

§3.15 Admin - Quarterly

- [] WHATSAPP_TRIGGERS tuning meeting - review with Logistics, adjust per-event toggles
 - [] SOP review - update this document with any process changes
 - [] Database backup integrity test - restore to a sandbox + run bug_check.py
-

§4. Decision Trees

§4.1 DN approval - Approve vs Reject vs Reschedule

```

                ?????????????????????
                ?  DN in HOD approval    ?
                ?  queue (or Logistics)   ?
                ?????????????????????
                    ?
            ?????????????????????????
            ?  Inspect: qty, family,      ?
            ?  lot, expiry, vehicle, driver ?
            ?????????????????????????
                    ?
            ?????????????????????????
            ?                               ?
Everything OK   Date wrong but content   Content wrong
            ?           is fine           ?
            ?           ?                 ?
            ?           ?                 ?
            [OK] Approve   ? Reschedule           [X] Reject
                           (set new date)         (must include reason)
                           ?                     ?
                           ?                     ?
                           Logistics decides      Warehouse rebuilds DN;
                           within 2h business    loop continues

```

Approve when: Qty matches PR - Family correct - Lot has > 90d shelf life - Vehicle + driver populated - No HSE concern with this delivery date

Reschedule (don't reject) when: Content is fine but YOU can't receive on the scheduled date (site shutdown, manpower, weather). Reschedule is faster than reject + rebuild.

Reject when: Qty doesn't match PR - Wrong material - Lot expiry inadequate - Wrong destination site (misroute) - Driver/vehicle missing critical info

§4.2 Vendor return vs Stock adjustment

```

????????????????????????????????
? Discrepancy discovered ?
????????????????????????????????
?
????????????????????????????????
? Was material RECEIVED already ?
? via a DN this site approved? ?
????????????????????????????????
?
????????????????????????????????
? ?
YES (in receipts) NO (still at warehouse)
? ?
? ?
????????????????????????????????
? Is the issue with ? ? Warehouse handles ?
? THIS DN (defective, ? ? internally - Return ?
? wrong qty), or ? ? to vendor flow ?
? general stock ? ? (Tab 8 Warehouse) ?
? count drift? ? ?????????????????????????
????????????????????????????????
?
????????????????????????????????
? ?
This DN General drift
? ?
? ?
<-? Vendor Stock Adjustment
Return (SK Entry Log ->
(any role) Stock Count tab)

```

Use Vendor Return when: A specific DN brought defective / wrong qty / quality-issue material. Reopens the originating PO line so Logistics can chase vendor.

Use Stock Adjustment when: Cycle-count discovers shelf qty ? system qty for a general reason (damage in storage, miscount, expired-disposal, etc.). Doesn't touch the PO chain.

§4.3 30-day return window - Override or not?

```

????????????????????????????????
?  SK / HOD wants to return  ?
?  material to logistics      ?
????????????????????????????????
?
????????????????????????????????
? Received in last 30d?      ?
????????????????????????????????
?
????????????????????????????????
?                               ?
YES                               NO
?                               ?
?                               ?
Standard return flow             Override required:
(SK Entry Log -> Return)         tick "Override 30-day window"
                                + write justification
                                ?
                                ?
                                HOD reviews - override
                                rows highlighted RED
                                ?
                                ??????????????????
                                ?           ?
                                Justified   Not justified
                                ?           ?
                                ?           ?
                                [OK] Approve      [X] Reject
                                           (use Stock Adjustment
                                           with reason `damaged`
                                           or `expired_disposal`)

```

§4.4 Force-close decision

```

????????????????????????????
? Logistics considering      ?
? force-closing PR/PO/line  ?
????????????????????????????
?
????????????????????????????
? Has stakeholder been      ?
? consulted (Site HOD)?     ?
????????????????????????????
?
NO ?????????????????????? YES
? ?
? ?
STOP - call HOD first.      Proceed to next gate.
Force-close is permanent.
?
????????????????????????????
? Is the reason in          ?
? scope of force-close?     ?
????????????????????????????
?
In scope ?????????????????? Out of scope
? ?
? ?
[OK] Force-close           Use alternative:
with full reason           - Vendor cancellation -> close PR
                           - Wrong qty on PO -> vendor return + new PO
                           - Wrong material -> vendor return
                           - Project cancelled -> close PR

```

In scope for force-close: Project cancellation - Vendor permanently unable to fulfill - PR raised in error and beyond email-recall - Item discontinued - Compliance violation

Out of scope (use other tool): Quantity / quality issues (use vendor return) - Date slippage (use reschedule) - Wrong destination site (rebuild DN, no force-close needed)

§5. Escalation Matrix

When a process step doesn't happen on time, here's who pings whom, when, and on which channel.

§5.1 PR -> PO escalation

Trigger	Wait time	Action	Channel
PR submitted to Logistics, no PO after 4 business hours	4h	Site HOD -> Logistics lead	WhatsApp
^ Still nothing after another 4h	8h	Site HOD -> Admin	App notification + email

Trigger	Wait time	Action	Channel
^ Still nothing after 24h	24h	Admin overrides - assigns alternate Logistics user	Admin Portal -> ? Users

§5.2 PO -> Assignment escalation

Trigger	Wait time	Action	Channel
PO issued, no warehouse assignment after 8 business hours	8h	Site HOD -> Logistics (with PO number)	WhatsApp
Assignment created, Warehouse hasn't acknowledged in 24h	24h	Logistics -> Warehouse Lead	WhatsApp + email
^ Still no ack after 48h	48h	Warehouse Lead -> Admin	Admin escalation

§5.3 DN approval escalation

Trigger	Wait time	Action	Channel
DN sitting at pending_logistics for > 4 business hours	4h	Warehouse -> Logistics	WhatsApp
DN sitting at pending_hod for > 8 business hours	8h	Logistics -> Site HOD	WhatsApp
DN sitting at pending_sk for > 24 hours after expected arrival	24h	HOD -> SK + Warehouse	WhatsApp

§5.4 Reschedule escalation

Trigger	Wait time	Action	Channel
Reschedule pending > 2 business hours during work day	2h	Requester -> Logistics	WhatsApp
^ Pending > 24h	24h	Site HOD -> Admin	App + email

§5.5 Vendor return escalation

Trigger	Wait time	Action	Channel
Vendor return raised, no acknowledgement from vendor in 7 days	7d	Logistics -> Vendor directly (email)	Email
^ Still unresolved after 14d	14d	Logistics -> Site HOD + Admin	App + email
Resupply date passed			

without delivery

T+1d

Logistics -> Admin

Critical

§5.6 Force-closure escalation

Force-closures fire critical notifications to Admin + originating Site HOD instantly. There's no escalation needed - the action is already at maximum severity. Admin reviews weekly per §3.13.

§6. Recovery Procedures

§6.1 DN rejected at HOD - how Warehouse recovers

1. Bell shows rejection notification with HOD's reason
2. Warehouse user reads the reason from the notification card (full text in the related DN row in `delivery_notes.rejection_reason`)
3. Determine what changed:
 - If qty wrong: prepare a new DN with corrected qty (the rejected DN's qty frees back into the available calculation immediately)
 - If material wrong: file an internal Stock Adjustment if you over-counted at receive, OR raise a vendor return if the wrong material was delivered to you
 - If date wrong: prepare new DN with the corrected date; HOD will approve in same cycle
7. Submit new DN -> Logistics -> HOD; loop continues

§6.2 Warehouse short-receives from vendor

1. Receive Goods tab - type the actual received qty (less than ordered)
2. System auto-flips `po_items.line_status` to `partially_delivered`
3. Add a note in the assignment notes describing the shortfall
4. Notify Logistics via WhatsApp with the PO number + shortfall qty
5. Logistics raises a Vendor Return with reason "Vendor under-shipped" (uses the same return flow, reopens the line for resupply)
6. New shipment from vendor -> Warehouse receives the remainder -> line flips to delivered

§6.3 Vendor return + resupply cycle

1. Issue surfaces (SK at site OR Warehouse at receive)
2. Determine raiser:
 - Issue at site after SK confirmed -> SK raises to HOD -> HOD raises to Warehouse via internal return -> Warehouse raises Vendor Return
 - Issue at warehouse before DN ship -> Warehouse raises Vendor Return directly
5. Logistics tracks return in <-? Vendor Returns tab
6. Expected_Resupply date set (default today + 14)
7. T-2 / T-1 / T-0 reminders fire on Expected_Resupply
8. When resupply arrives: Warehouse re-receives -> line `Delivered_Qty` recalculates -> status moves forward

§6.4 HOD absent for > 24h with pending DN

1.

Admin shadows HOD Portal (admin can access via the existing override)

2. Admin reviews the pending DN(s) in HOD Portal -> ? DN Approvals
3. Admin approves on behalf, leaves a note in the DN attesting to the cover
4. Audit log records DN_HOD_APPROVE with admin's username
5. HOD-on-return is notified to review post-fact

§6.5 Logistics absent - Admin override

1. Admin enters Logistics Portal via shadow access
2. Admin acts as Logistics for the day (approve PRs, issue POs if pre-arranged, approve reschedules)
3. Audit log records each action with admin's username (not the absent Logistics user)
4. Admin notifies the absent Logistics user via WhatsApp before signing off

§6.6 Force-closure made in error

There's no "undo force-close" button (yet - on the v3.0 backlog). Workaround:

1. Admin -> Master DB Editor -> relevant table (purchase_orders / pr_master)
2. Edit status from force_closed back to open (PO) or remove the logistics_status='force_closed' flag (PR)
3. Add a corresponding po_force_closures row with reason "Reversed in error - see audit"
4. The Site HOD's "Force-closures affecting me" tab will show both the original close + the reversal

Use sparingly - this is invasive. Better: communicate the reversal verbally and update via DB Editor with a paper trail.

§6.7 RL/BL DN was prepared as combined (system rejected, user confused)

1. UI shows: "Strict separation violated: this DN spans multiple RL/BL families. Prepare one DN per family."
2. Warehouse user reduces Ship Qty to 0 on either the RL lines OR the BL lines (not both)
3. Save DN draft - succeeds with only one family
4. Prepare second DN with the other family - same source PO, different family
5. Submit both DN's separately; both flow through approval independently

§7. Quick Reference Cards

§7.1 Site HOD - One-page summary

Daily must-dos: Bell badge clear - Pending Receipts approved - DN Approvals queue empty - EOD commit by 17:00 - Sign out clean. Submit PR to Logistics: HOD Portal -> PRs -> ? Submit PR(s) to Logistics expander -> multi-select -> Submit. See what's incoming: ? In-Transit -> Active in-transit (sorted by closest-to-SK). Request reschedule: ? In-Transit -> click DN card -> ? Request reschedule (date defaults to ETA + 3 days). Escalate stuck PR: WhatsApp Logistics if no PO after 4h business hours. Force-closure questions: Check ? In-Transit -> Force-closures sub-tab; escalate to Admin if reason looks weak.

§7.2 Logistics - One-page summary

Hourly must-dos: Bell badge ? 30min - Incoming PRs queue ? 4h - ? Reschedules ? 2h. Issue PO from PR: Incoming PRs -> drill in -> ? Use this PR to create a PO -> fill header -> save. Issue PO from vendor PDF: ? Create PO -> ? PDF upload -> Extract -> Review -> Save. Assign to warehouse: ? Assign -> pick PO + WH -> confirm items -> Expected Delivery -> Assign. Approve DN date: ?? Outbound DN queue (Tab 4 visible on Warehouse Portal - Logistics actions through ?? DN approval queue) -> [OK] Approve. Force-close: Always call Site HOD first. Then ? Force-Close with detailed reason.

§7.3 Warehouse User - One-page summary

Per shift: Bell badge clear - Incoming Assignments acked within 1h - Outbound DN all submitted (no orphan drafts). Receive goods: Receive Goods -> assignment -> type Receive Now per line -> Record receipt. Prepare DN: Prepare DN -> pick PO + site -> Ship Qty per line -> header -> Save -> ?? Submit. Strict rule: ONE family (RL or BL) per DN. System rejects mixed. Returns from site: <-? Returns from Site -> DN -> return qty + reason -> <-? Raise to vendor.

§7.4 Store Keeper - One-page summary

On delivery day: Entry Log -> Receipt Staging -> ? Incoming DN expander -> inspect -> [OK] Mark as Received. On every day: Consumption Log -> record issues; Receipt Staging for direct deliveries; Stock Count for cycle counts; QR Label Request for new labels. Don't: Mark received without physical inspection. Mix Returnable (tool loans) with Return Items (real returns). Smart Scan - when CV detection is wrong: the green "Auto-filled" card can be edited before submit. If a yellow "Top candidates" radio appears, pick the right one or override the tool name in the manual form. If no candidates show (or no active model), just type it manually - borrower fields stay pre-filled from the badge scan. The submitted loan records which path (CV vs manual) was used so adoption telemetry stays honest. Full workflow: USER_MANUAL.md §4.5.0. Trainer guide: docs/cv_training_guide.md.

§7.5 Admin - One-page summary

Weekly: ? Logistics Oversight scan; force-closures review; user registrations approve. Monthly: 3 procurement reports (PO Status / Warehouse Throughput / Force-Closures); bell-noise audit. Quarterly: SOP review; backup integrity test. Override: Use Master DB Editor with a paper trail; document every override in system_audit_log via the affected helper.

§8. Glossary

For complete vocabulary (FEFO, EOD, PR, DN, OCR, WAL, etc.) see USER_MANUAL.md §11.5. Procurement-chain-specific terms only here:

Term	Meaning
Procurement chain	The in-app SQL-driven workflow from Site PR -> Logistics PO -> Warehouse DN -> Site SK receipt. Replaces (optionally) the legacy email/PDF flow.
DN state machine	The seven-step lifecycle of a Delivery Note (draft ->

pending_logistics -> logistics_approved -> pending_hod
-> hod_approved -> pending_sk -> received), with
rejected as terminal from any pending...

Term	Meaning
RL/BL strict separation	Hard rule that Rubber Lining and Brick Lining never share a PO group, DN, or warehouse aggregation. Enforced at three layers.
Warehouse-blind pricing	Hard rule that warehouse_user role never sees Unit_Price, Total_Price, or monetary header fields on any PO. Three enforcement layers.
Logistics shadow	Admin's read+write access to the Logistics Portal as if they were a Logistics user. Used for absence cover.
Warehouse shadow	Admin's access to the Warehouse Portal with a sidebar warehouse picker. Used for absence cover or audit.
Force-closure	Logistics' one-way close of a PR / PO / line with mandatory reason. Notifies Admin + Site HOD with critical severity.
T-2 / T-1 / T-0 reminders	Daily sweep job firing notifications 2 days before, 1 day before, and on the expected delivery date. Idempotent.
App notification	A row in app_notifications, surfaced via the sidebar bell. Always fires; in-app only.
WhatsApp trigger toggle	config.WHATSAPP_TRIGGERS[event_key] = True/False to per-event control WhatsApp emission. In-app is unaffected.

§9. Change Log

Version	Date	Author	Changes
1.0	2026-06	Initial release	First publication alongside Hub v3.0 procurement chain. Covers Site HOD, Logistics, Warehou...

End of SOP. Review quarterly. Update on any role change, vendor onboarding, escalation-time tweak, or DN-state-machine alteration. Companion: USER_MANUAL.md (UI reference) - handoff.md (technical architecture).